Worksheet 1

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1 T-test

1.1 The t-test assumes normally distributed data. For an increasing number of samples N = {5,10,20,50,100}, sample from each of these distributions:

```
set.seed(42)
library(ggplot2)
sample_sizes <- c(5,10,20,50,100)
n_reps <- 1000
alpha <- 0.05
results <- list()
```

working on t test, next chunk sets distributive functions

```
uniform_cont <- function(n){
  runif(n, -1, 1)
}
students_t <- function(n){
  rt(n, df=1)
}
uniform_discrete <- function(n){
```

```
  sample(c(-1, 1), size = n, replace = TRUE)
}
```

1. continuous uniform distribution in [-1, 1]
2. Student-t Distribution ($= 1$)
3. Discrete Uniform Distribution $\{-1, 1\}$

```
# Initialize a data frame to store all p-values
all_pvals <- data.frame()

for (n in sample_sizes) {
  for (i in 1:n_reps) {
    # Sample and Perform two-sided t-tests, mean=0
    uc_sample <- uniform_cont(n)
    p_uc <- t.test(uc_sample, mu = 0)$p.value

    st_sample <- students_t(n)
    p_st <- t.test(st_sample, mu = 0)$p.value

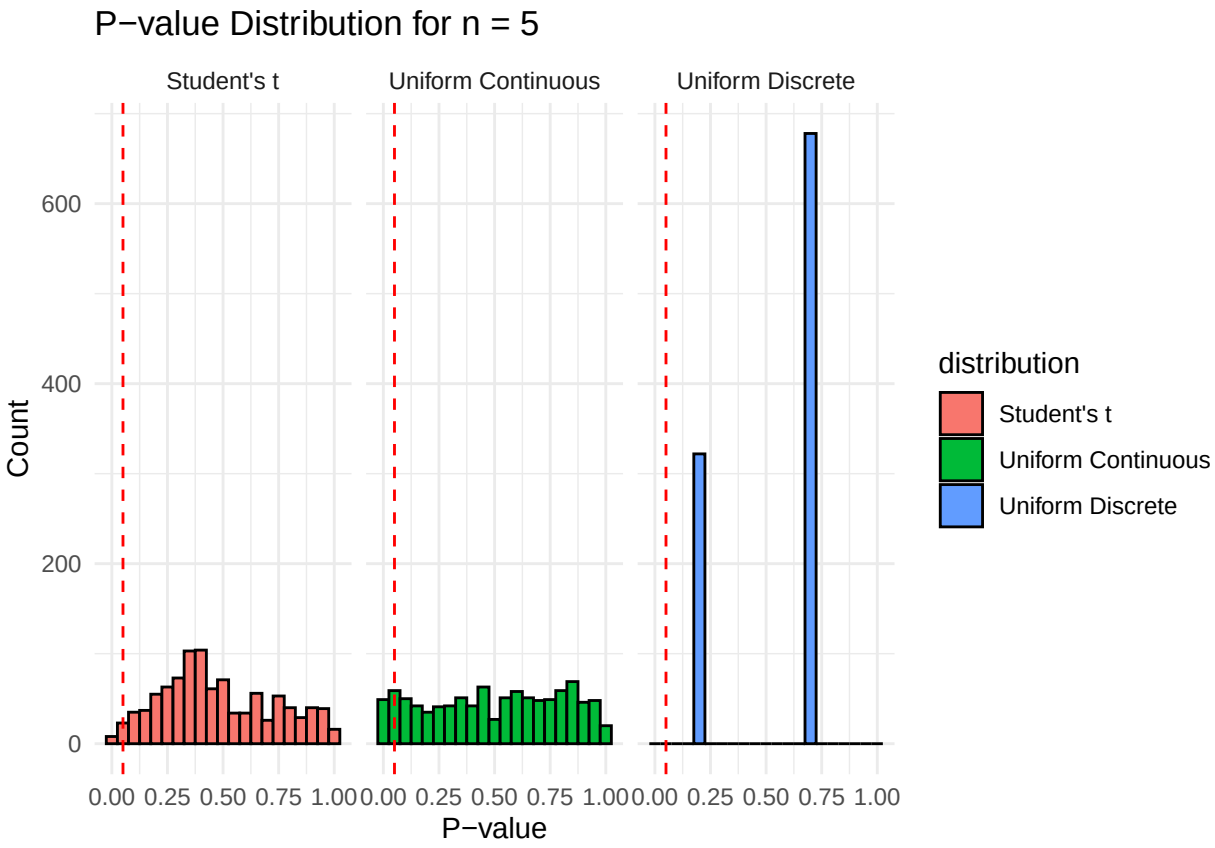
    # Unif discrete with resampling if t-test fails
    success <- FALSE
    while (!success) {
      ud_sample <- uniform_discrete(n)
      test_result <- try(t.test(ud_sample, mu = 0), silent = TRUE)
      if (!inherits(test_result, "try-error")) {
        p_ud <- test_result$p.value
        success <- TRUE
      }
    }

    all_pvals <- rbind(all_pvals, data.frame(
      n = n,
      rep = i,
      p_value = p_uc,
      distribution = "Uniform Continuous"
    ))
    all_pvals <- rbind(all_pvals, data.frame(
      n = n,
      rep = i,
      p_value = p_st,
      distribution = "Student's t"
    ))
    all_pvals <- rbind(all_pvals, data.frame(
      n = n,
      rep = i,
      p_value = p_ud,
      distribution = "Uniform Discrete"
    ))
  }
}
```

Plot 1: P-value distr for sample_size=5

```r
library(ggplot2)

pvals_n5 <- subset(all_pvals, n==5)
ggplot(pvals_n5, aes(x = p_value, fill = distribution)) +
  geom_histogram(binwidth = 0.05, position = "dodge", color = "black") +
  geom_vline(xintercept = alpha, color = "red", linetype = "dashed") +
  facet_wrap(~distribution) +
  labs(title = "P-value Distribution for n = 5",
       x = "P-value",
       y = "Count") +
  theme_minimal()
```



Only student's t looks like a uniform distribution between 0 and 1 Count p-values below 0.05 for each distribution

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
```

```
##
##     intersect, setdiff, setequal, union
```

```r
significant_ratios <- all_pvals %>%
  group_by(distribution) %>%
  summarise(
    total = n(),
    below_alpha = sum(p_value < alpha),
    ratio = below_alpha / total
  )
```

Significant p-value Ratios of Distributions:

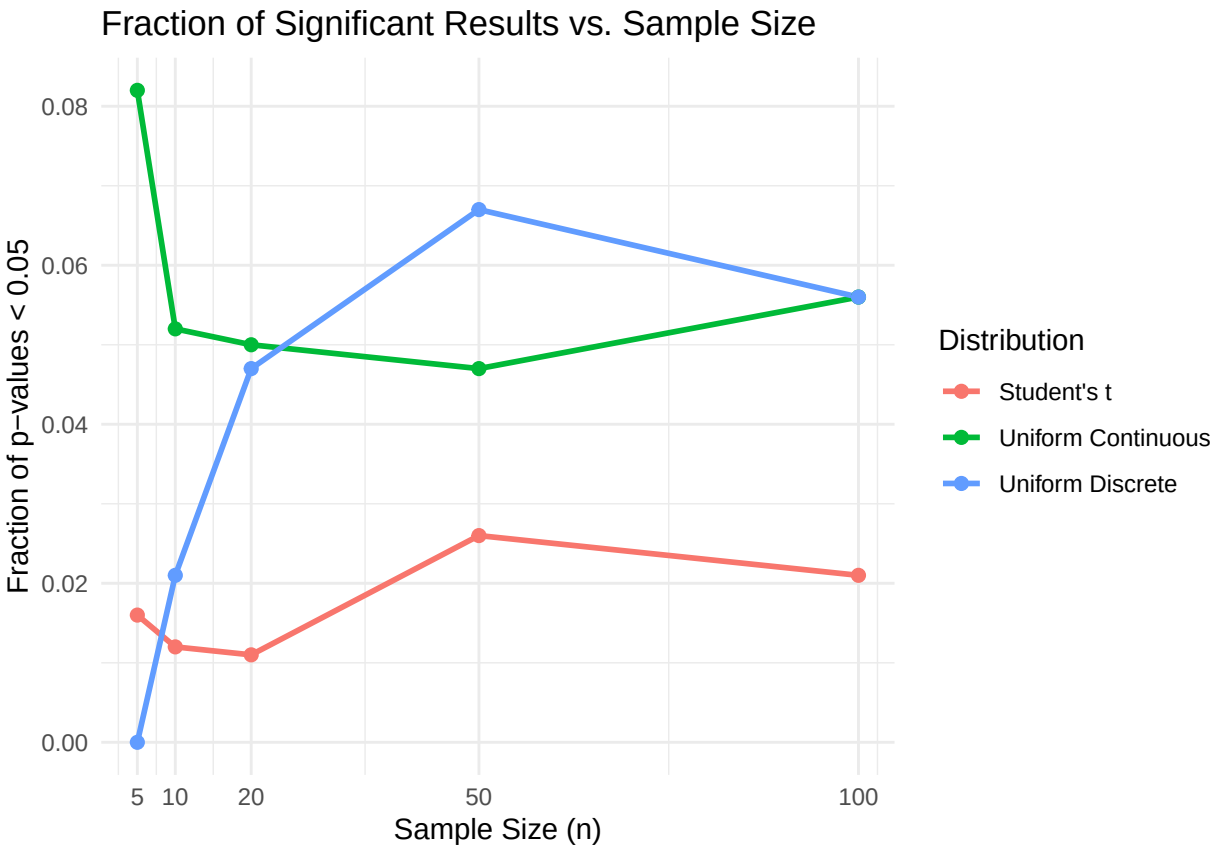
- Uniform continuous: $287/5000 = 0.0574$ S
- Student's t: $86/5000 = 0.0172$
- Uniform discrete: $191/5000 = 0.0382$

Computing fraction of significant p-values by sample size and distribution

```r
fraction_df <- all_pvals %>%
  group_by(distribution, n) %>%
  summarise(
    total = n(),
    below_alpha = sum(p_value < alpha),
    fraction_significant = below_alpha / total,
    .groups = "drop"
  )
```

Plot 2: the fraction of significant results as the sample size increases

```r
ggplot(fraction_df, aes(x = n, y = fraction_significant, color = distribution)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = sample_sizes) +
  labs(
    title = "Fraction of Significant Results vs. Sample Size",
    x = "Sample Size (n)",
    y = "Fraction of p-values < 0.05",
    color = "Distribution"
  ) +
  theme_minimal()
```



1.2 Interpretation:

- Student's t distribution has the lowest fraction of significant p-values due to heavier tails of the student's t distribution with df=1 and its infinite standard deviation.
- The denominator of the t-statistic (which is the estimated standard error) gets inflated. So the t-statistic is smaller (closer to 0 than it should be). As a result even though our mean was supposed to be close to 0, we get an extreme result of significant p-value ratios around 0.02 instead of 0.05.
- Because the t-statistic is small (due to large s.e), the p-values tend to be larger than they should be.
- Smaller absolute t-statistic -> interpret that sample mean is close to 0 -> p-value larger.
- To sum it up, t-test can be irrelevant for distributions with very high std deviation such as Student's t distribution.

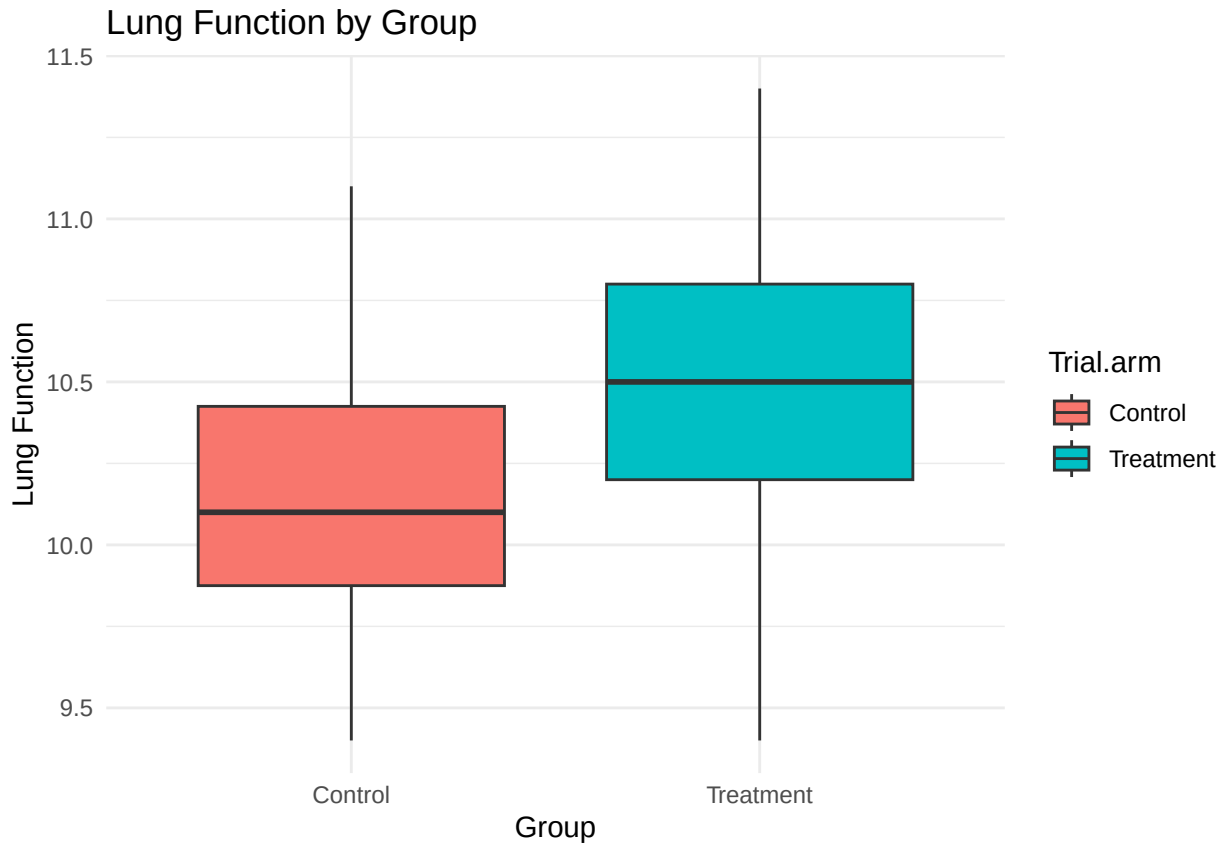
2 T-tests

2.1 A small clinical trial on asthma patients has been run measuring the lung function of a control group on a placebo and a treatment group on a new drug.

```r
lung_data <- read.csv("/Users/payalpriyadarshini/Documents/Data Science_2nd sem/lung_data.csv")
head(lung_data)
```

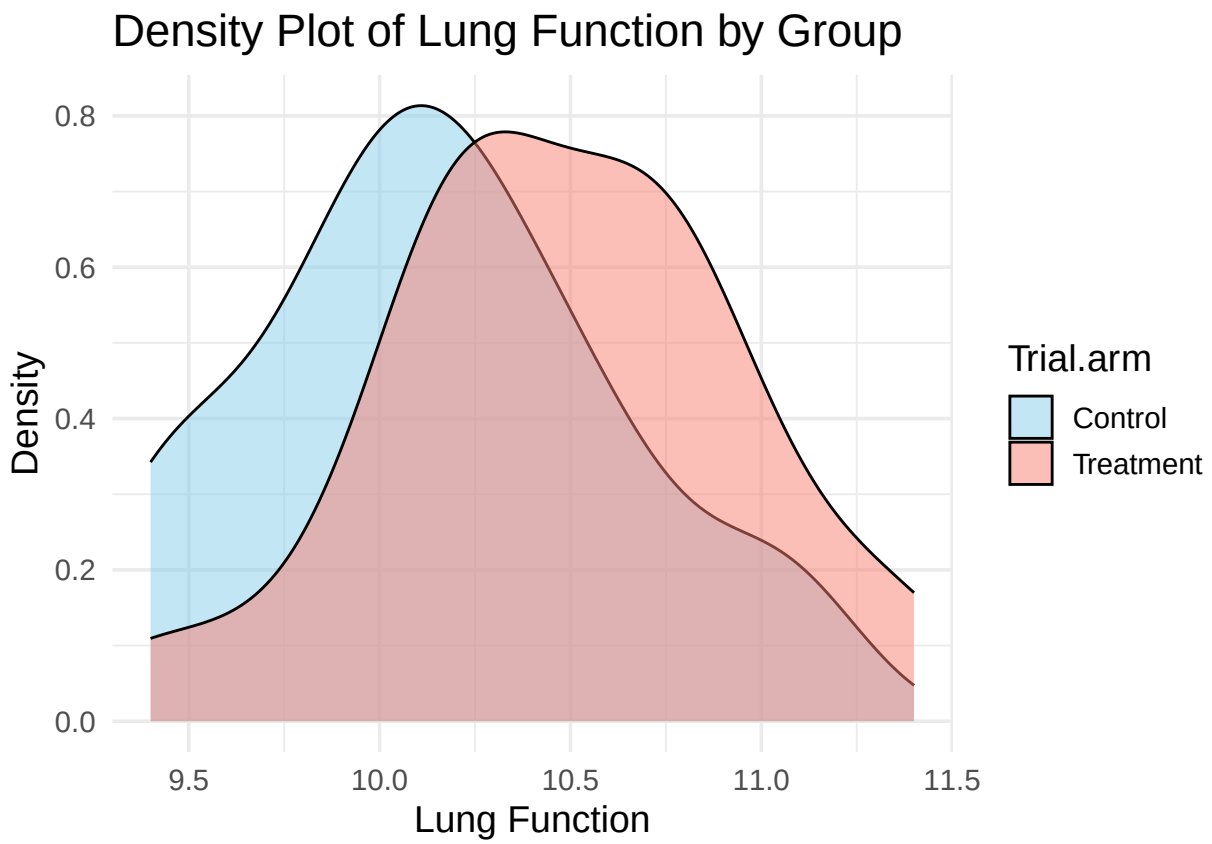
```
##   Subject.id Lung.function Trial.arm
## 1          1          11.0   Control
## 2          2          11.1   Control
## 3          3           9.5   Control
## 4          4          10.1   Control
## 5          5           9.7   Control
## 6          6           9.4   Control
```

```r
ggplot(lung_data, aes(x = Trial.arm, y = Lung.function, fill = Trial.arm)) +
  geom_boxplot() +
  labs(title = "Lung Function by Group", x = "Group", y = "Lung Function") +
  theme_minimal()
```




```r
ggplot(lung_data, aes(x = Lung.function, fill = Trial.arm)) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Density Plot of Lung Function by Group",
    x = "Lung Function",
    y = "Density"
  ) +
```

```
  scale_fill_manual(values = c("skyblue", "salmon")) +
  theme_minimal(base_size = 14)
```




```
# Histogram
ggplot(lung_data, aes(x = Lung.function, fill = Trial.arm)) +
  geom_histogram(binwidth = 0.4, color = "black") +
  facet_wrap(~ Trial.arm, ncol = 2) + # Create separate panels for each Trial.arm
  labs(title = "Distribution of Lung Measurements by Trial Arm",
       x = "Lung Function", y = "Count", fill = "Trial Arm") +
  theme_minimal()
```




```
library(dplyr)
t_test <- t.test(Lung.function ~ Trial.arm, data = lung_data)
print(t_test)
```

```
##
##  Welch Two Sample t-test
##
## data:  Lung.function by Trial.arm
## t = -2.1569, df = 37.988, p-value = 0.0374
## alternative hypothesis: true difference in means between group Control and group Treatment is not eq
## 95 percent confidence interval:
##  -0.63003538 -0.01996462
## sample estimates:
##   mean in group Control mean in group Treatment
##                  10.150                  10.475
```

2.2 Statistical Analysis Report: Lung Function in Asthma Clinical Trial

Objective- To assess whether there is a statistically significant difference in lung function between asthma patients receiving a placebo (control) and those receiving a new drug (treatment).

Data Visualization

Three visualizations were created:

1. Boxplot: Compares the lung function distributions between the control and treatment groups. The treatment group appears to have a slightly higher median lung function.
2. Histogram: Shows the distribution of lung function values within each group. It is observed that the treatment group has a bell shaped distribution and is symmetric and centered while the control group is slightly shifted to the right.
3. Density plot: The treatment group (salmon) has a rightward shift with broader spread, showcasing higher and variable lung function values. The control group (sky blue) is more peaked, with lung function values varying around a slightly lower range.

Statistical test (t-test)

A two sample t-test was conducted to compare the difference in the two independent groups and understand the significance of their difference. More specifically, a Welch's t-test was conducted as it is more robust for groups with different variances and sample sizes, and avoids presumption about the variance in the data. Results: We conducted a statistical comparison of the Lung function with the Trial arm(group), the following are the results obtained:

- t-statistic: -2.1569
- Degrees of freedom: 37.988
- p-value: $0.0374 < 0.05$
- 95% Confidence Interval: [-0.630, -0.019]
- Control Mean: 10.150
- Treatment Mean: 10.475

Interpretation:

1. P-value is less than the significance level of 0.05, hence there is a statistical difference in lung function of two groups.
2. We reject the null hypothesis and accept the alternate hypothesis of having significant difference in lung function of control and treatment.
3. From the density plot we can observe that the drug has a positive effect due to the right shift.
4. Non- zero C.I. suggests we have statistical evidence that the drug has a reliable, positive effect on lung function Mean value of Treatment is higher than control group
5. These findings support further clinical investigation and potential drug development.

3 Review questions

Mark True or False for each of the following statements:

- p-values are used to calculate the probability of the null hypothesis given the data. **FALSE**
- The significance level alpha used in statistical hypothesis testing is the probability of rejecting the null when it is true **TRUE**
- The Central Limit Theorem only holds if the population from which we are sampling is normally distributed. **FALSE**
- As the sample size gets larger, the standard error of the sampling distribution of the sample mean gets larger as well. **FALSE**